Week 1: Large Numbers, State Variables, and Heat

Ben Kyvik

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Disclaimer:

These notes are not to be sold or widely distributed. They are highly derivative of lecture content, which, in turn, draws heavily from the textbook (in this case *Concepts in Thermal Physics* by Stephen and Katherine Blundell). These notes are instead intended to help myself and my peers.

1 Large Numbers

Thermodynamics is concerned with the shenanigans of **big** systems. It seeks to model the bulk behavior of microscopic molecules within a macroscopic system. The challenge here is that these systems contain more molecules than you could reasonably expect to count, much less watch.

On the scale of everyday life, the behavior of a single particle doesn't affect you at all. If I stole one atom from you every day for the rest of your life, I promise you wouldn't miss it. Hence, we count molecules by the *mole*.

Much like a dozen eggs is 12 eggs, a **mole** of eggs is *Avogadro's number* of eggs, where:

$$N_A = 6.02214076 \times 10^{23}\,\text{mol}^{-1} = 602,214,076,000,000,000,000,000 \tag{1}$$

For perspective, that's roughly six hundred billion trillion eggs. These are the kind of quantities we're dealing with in thermal physics.

What was Avogadro on? Why this? Why can't we use a power of ten like 10^{24}. Well, it's a historical choice that relates our units of measurement. Namely, the mole was defined as the number of atoms in one gram of ^{12}C. This is the isotope of carbon with 6 protons and 6 neutrons, and if a proton and neutron are both roughly 1 atomic mass unit, then:

$$1\,\text{mol} = \frac{\text{amu}}{\text{g}} \tag{2}$$

Notice that this is a ratio of mass units, meaning that a mole has no units. It's just a count of things. This correspondence makes it easy to discuss the mass of a sample. The **molar mass** of something is the mass of one mole of it. For instance, one molecule of water has a mass of 18 amu, so it's *molar mass* is 18 g.

It's easy to show that, when dealing with a larger and larger sample size, the relative fluctuations shrink. Specifically, as the average value $\langle A \rangle$ grows in proportion to the number of particles N, then the total fluctuation δA grows in proportion to $\sqrt{N}$, and we find that the *relative* fluctuation actually *shrinks*:

$$\delta A_{relative} = \frac{\delta A}{A} \propto \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}} \tag{3}$$

And for a large-scale system of hella particles, this value becomes effectively zero:

$$\lim_{N\to\infty} \delta A_{relative} = 0 \tag{4}$$

When we deal with a system of this magnitude, the number of possible configurations are virtually endless. This begs interesting questions in combinatorics.

As an example, consider a system of n atoms, each with an energy of either 0 or 1. We might want to know how many configurations are possible for a total energy of k.

Where should we put the first 1? Well, there's n different atoms to assign it to, so n possibilities for $k = 1$. Where does the next one go? Now there's already one occupied atom, and $n - 1$ remaining choices. In general, each time we assign a 1, we are left with one less remaining choice. Multiplying the possible choices for each step, we have:

$$n(n-1)(n-2)...(n-k+1) \tag{5}$$

This looks an awful lot like the factorial function $n!$, where:

$$n! = n(n-1)(n-2)...(2)(1) \tag{6}$$

Notice we could write $n!$ as:

$$n! = n(n-1)...(n-k+1)(n-k)! \tag{7}$$

So we can get the desired product if we just divide by $(n-k)!$

$$n(n-1)...(n-k+1) = \frac{n!}{(n-k)!} \tag{8}$$

Finally remember that we really don't care about the possible configurations of energized atoms within k, so we also need to divide the number of configurations of k atoms:

$$\#\text{ of possible configurations} = \frac{n!}{k!(n-k)!}$$

This is what we call a k**-combination** of n elements, denoted ${}^kC_n$, or n **choose** k:

$$\binom{n}{k} = \frac{n!}{k!(n-k!)} \tag{9}$$

When dealing with large numbers, the factorial function infamously gets really big really quickly (much faster than the exponential function). $5! = 120$, $10! = 3.6288 \times 10^7$, $20! = 2.432902 \times 10^{18}$, and $100! \sim 10^{158}$. How many configurations of one mole? That's:

$$N_A! = (6.022 \times 10^{23})(6.022 \times 10^{23} - 1)...(2)(1)$$

To be honest, I'm kind of afraid to put that into my calculator, and frankly nobody should. Fortunately, we have a heuristic that approximates factorial growth with increasing precision for larger n. Specifically, we find that:

$$lnn! \approx n(\ln(n) - 1) \tag{10}$$

Logarithms are way easier to work with for large numbers, because they turn multiplication into addition, which is a lot less computationally intensive. We can use the exponential function to arrive back at our approximate $n!$

$$e^{\ln n!} = n! \approx e^{n(\ln(n)-1)} \tag{11}$$

2 State Variables

A **state variable** is defined (by my professor at least) as an average physical quantity used to describe a macroscopic system. If we think of a gas inside a box, the position and momentum of all molecules will average to zero, and hence these quantities tell us nothing of value.

Much more useful is the average energy of a molecule in the system, $\langle E_K \rangle$, which we'll see translates to the temperature T. Also important is the system's volume V and the number N of particles inside it.

We can split these state variables into two different types: extensive and intensive.

To illustrate this distinction, consider a box filled with particles. The box has a volume V, a temperature T, a pressure P, a total energy E_{total}, and N particles inside of it. (If I was getting paid for this, I might have drawn one here).

Now suppose we partition the box and break it in half. Now we have two boxes, each with volume $V' = V/2$ and containing $N' = N/2$ particles. Likewise, the total energy is now $E'_{total} = E_{total}/2$. This means that the volume and particle count scale with the system.

If we consider instead the *temperature* and *pressure*, we find no change! The new pressure is $P' = P$ and the temperature is $T' = T$, as before!

We call the total energy, volume, and particle count **extensive** properties: properties that scale with the system. Meanwhile, the temperature and pressure are **intensive**, since they remain constant as the system's size changes.

3 Ideal Gas Law

A confined gas with n moles of particles has 3 degrees of freedom: pressure P, volume V, and temperature T. The natural question here is how these three properties relate to each other. If you keep a first quantity constant and vary the second, how will the third property change?

The first of these relations was discovered in 1662 by Robert Boyle, who fixed the temperature of a gas and found that pressure is inversely proportional to volume. Hence **Boyle's Law**:

$$P \propto \frac{1}{V} \tag{12}$$

Jacques Charles was posthumously credited with the discovery of **Charles' Law** some time in the 1780s, which found that, at constant pressure, the volume of a gas grows in proportion to its temperature:

$$V \propto T \tag{13}$$

The last person to cash in on this question was Joseph-Louis Gay-Lussac, who kept the volume fixed and saw that pressure is proportional to temperature, and so we have **Gay-Lussac's Law**:

$$P \propto T \tag{14}$$

Frankly, there's no point in memorizing these three relations or who discovered what, because it's all really redundant. For the sake of efficiency, we can combine all three into a single, holy grail equation for a gas, the **ideal gas law**:

$$\boxed{PV = nRT} \tag{15}$$

where n is the number of moles of gas and R is the **universal gas constant**:

$$R = N_A k_B \approx 8.314\,\mathrm{J\ mol^{-1}\ K^{-1}} \tag{16}$$

where $k_B \approx 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ is the Boltzmann constant which we haven't learned yet.

4 Heat

Heat, (denoted Q) is defined as the *transfer of thermal energy*. Hence, kind of like work (the change in kinetic energy), heat has units of energy, and is measured in Joules. Similarly, the rate of heat has units of *power*, and is measured in Joules per second, or *Watts*.

As a general rule, heat likes to spread out as much as possible, flowing from hotter to colder environments until some kind of thermal equilibrium is reached. However, heat doesn't affect every material the same way. Some materials require much more heat than others for the same increase in temperature. So we define a property called **specific heat**, the extent of change in temperature for a given amount of heat:

$$\text{Heat capacity} = \frac{dQ}{dT} \tag{17}$$

Divide by the mass of the material, and we get the **specific heat**, C:

The specific heat of a system depends on the constraints applied to it. In the spirit of the ideal gas law, suppose we fix the volume:

$$C_V = \left.\frac{\partial Q}{\partial T}\right|_V \tag{18}$$

And what if we fix the pressure?

$$C_P = \left.\frac{\partial Q}{\partial T}\right|_P \tag{19}$$

C_V and C_P are the **heat capacity at constant volume** and **heat capacity at constant pressure**, respectively.

In general, $C_V < C_P$, since C_P allows the system to expand, and requires additional heat to make up for the energy spent on expansion.